WORKED SOLUTION

STEP III 2003, Question 6

Pure • Telescoping Trigonometric Sums

STEP III 2003, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q6

The key idea

A sum of cosines whose angles go up in equal steps is awkward as it stands, but multiplying every term by $2\sin\frac{1}{2}\theta$ turns each cosine into a *difference* of two sines, and the sum then telescopes. What is left is a difference of two sines, which the factor formula turns into a product. Setting a product to zero is easy. The one thing to watch is that we multiplied by $\sin\frac{1}{2}\theta$, which can itself be zero.

Solution

The identity

Using $\sin(A+B)-\sin(A-B)=2\cos A\sin B$ with $A=r\theta$ and $B=\frac{1}{2}\theta$:

$$\sin\left(r+\tfrac{1}{2}\right)\theta-\sin\left(r-\tfrac{1}{2}\right)\theta=2\cos r\theta\,\sin\tfrac{1}{2}\theta,$$

which is the required result.

Summing the series

Write

$$S(\theta)=\sum_{r=a}^{b-1}\cos r\theta.$$

Multiplying by $2\sin\frac{1}{2}\theta$ and using the identity for each r from a to $b-1$,

$$\begin{aligned}2\sin\tfrac{1}{2}\theta\;S(\theta)&=\sum_{r=a}^{b-1}\left[\sin\left(r+\tfrac{1}{2}\right)\theta-\sin\left(r-\tfrac{1}{2}\right)\theta\right]\\&=\sin\left(b-\tfrac{1}{2}\right)\theta-\sin\left(a-\tfrac{1}{2}\right)\theta,\end{aligned}$$

because each $\sin\left(r+\frac{1}{2}\right)\theta$ cancels the $-\sin\left((r+1)-\frac{1}{2}\right)\theta$ from the next term, leaving only the top end of the last term and the bottom end of the first.

Now we use $\sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$, with $\frac{P+Q}{2} = \frac{1}{2}(a+b-1)\theta$ and $\frac{P-Q}{2} = \frac{1}{2}(b-a)\theta$:

$$\sin\tfrac{1}{2}\theta\ S(\theta) = \cos\tfrac{1}{2}(a+b-1)\theta\ \sin\tfrac{1}{2}(b-a)\theta. \qquad (*)$$

Solving $S(\theta) = 0$

? **Ask yourself** Can we just divide $(*)$ by $\sin\frac{1}{2}\theta$?

Only when $\sin\frac{1}{2}\theta \neq 0$, that is, when θ is not a multiple of 2π. So we deal with those values first. If $\theta = 2m\pi$ for an integer m, then every term $\cos r\theta$ equals 1, so $S(\theta) = b - a \geq 2$. These values are never solutions.

For every other θ we have $\sin\frac{1}{2}\theta \neq 0$, and by $(*)$,

$$S(\theta) = 0 \iff \cos\tfrac{1}{2}(a+b-1)\theta = 0 \ \text{ or } \ \sin\tfrac{1}{2}(b-a)\theta = 0.$$

- $\cos\frac{1}{2}(a+b-1)\theta = 0$ exactly when $\frac{1}{2}(a+b-1)\theta = \left(k+\frac{1}{2}\right)\pi$, that is,

$$\theta = \frac{(2k+1)\pi}{a+b-1}, \qquad k \in \mathbb{Z}.$$

  (Note $a+b-1 \geq 3$, so we are not dividing by zero.) None of these is a multiple of 2π: that would need $2k+1 = 2m(a+b-1)$, an odd number equal to an even one. So all of them are solutions.

- $\sin\frac{1}{2}(b-a)\theta = 0$ exactly when $\frac{1}{2}(b-a)\theta = k\pi$, that is,

$$\theta = \frac{2k\pi}{b-a}, \qquad k \in \mathbb{Z}.$$

  This is a multiple of 2π exactly when $b-a$ divides k, and we have already seen that those values fail. So we must exclude them.

? **Ask yourself** Why must the multiples of 2π be thrown out of the second family, when $(*)$ seems to be satisfied there?

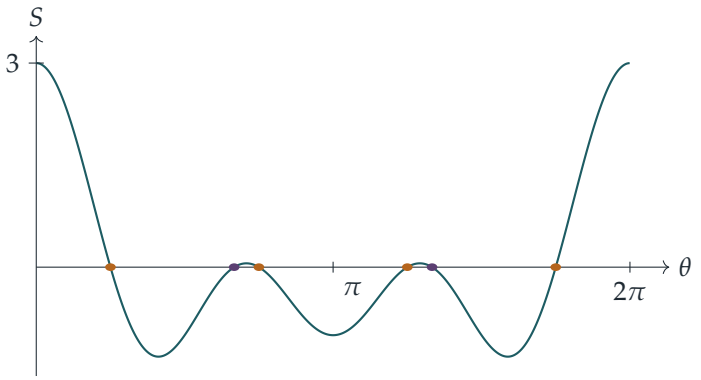
At $\theta = 2m\pi$ both sides of $(*)$ are zero whatever S is, so $(*)$ tells us nothing about S there. It was a false lead created by multiplying through by $\sin\frac{1}{2}\theta$. Since $b-a \geq 2$, the second family still contains genuine solutions, such as $\theta = 2\pi/(b-a)$.

Putting this together, the solutions are exactly

$$\boxed{\theta = \frac{(2k+1)\pi}{a+b-1}\ (k \in \mathbb{Z}) \qquad \text{or} \qquad \theta = \frac{2k\pi}{b-a}\ (k \in \mathbb{Z},\ b-a \nmid k).}$$

The two families can overlap. That does no harm, since a value in both is still just one solution.

For example, with $a = 1$ and $b = 4$ the equation is $\cos\theta + \cos 2\theta + \cos 3\theta = 0$. In $(0, 2\pi)$ its roots are $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ from the first family and $\frac{2\pi}{3}, \frac{4\pi}{3}$ from the second. Note that $\theta = 0$ and $\theta = 2\pi$, where $S = 3$, are left out.



The curve $S = \cos\theta + \cos 2\theta + \cos 3\theta$: the amber roots come from the cosine factor and the plum roots from the sine factor.

Checks

- With $a = 1$, $b = 4$: at $\theta = \frac{\pi}{4}$, $\frac{\sqrt{2}}{2} + 0 - \frac{\sqrt{2}}{2} = 0$, and at $\theta = \frac{2\pi}{3}$, $-\frac{1}{2} - \frac{1}{2} + 1 = 0$.
- Numerically, $(*)$ holds at random θ for many pairs a, b. For every $1 \le a \le 6$, $a + 2 \le b \le 11$, a fine scan of S on $(0, 6\pi)$ finds exactly the roots given by the formula and no others. For instance, with $a = 1$, $b = 3$ the root $\theta = 13\pi/3 \approx 13.6136$ comes from the first family.

Common slips

- Including $\theta = 2m\pi$ by dividing $(*)$ by $\sin\frac{1}{2}\theta$ without checking; there $S = b - a \neq 0$.
- Getting the end terms of the telescoping sum wrong: they are $\sin\left(b - \frac{1}{2}\right)\theta$ and $\sin\left(a - \frac{1}{2}\right)\theta$, because the last cosine is $\cos(b-1)\theta$.
- Giving only solutions in one interval such as $[0, 2\pi)$, when the question asks for all of them.

Beyond STEP

The closed form for $\sum \cos r\theta$ is the real part of a geometric series in $e^{i\theta}$. With $a = -n$ and $b = n + 1$ it gives the *Dirichlet kernel* $\sin\left(n + \frac{1}{2}\right)\theta / \sin\frac{1}{2}\theta$, which is at the heart of the theory of Fourier series and of whether they converge. Needham's *Visual Complex Analysis* is a good place to see why trigonometric sums like this become simple once you work with $e^{i\theta}$.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.